

Lab 2 Worksheet — Mobile Display Technology and Digitizer Fault Diagnosis

Name: ____ Date: ____

Exam objective: Compare LCD, IPS, VA and OLED panel technologies and correctly distinguish a display panel fault from a backlight, inverter or digitizer fault (Core 1 objective 1.2).

Record as you go

Step	What you did	What you observed
1	Build a comparison table with rows for TN, IPS, VA and OLED and columns for backlight required, colour accuracy, viewing angle, response time, contrast ratio and typical use.	
2	Shine a torch at a dark area of a powered-off LCD panel and then an OLED panel, and record which one reflects a visible backlight layer.	
3	Display a full-black image on both an LCD and an OLED screen in a dark room and record the difference in black level, then explain it from the backlight architecture.	
4	For the symptom 'image is visible only under a bright torch', identify the failed component and justify it — the panel is working, so the backlight or inverter has failed.	
5	For the symptom 'image is perfect but touch does not respond anywhere', identify the failed component as the digitizer and note that on bonded assemblies it is replaced with the panel.	
6	For the symptom 'flickering that worsens as the machine warms up on an older CCFL laptop', identify the inverter as the probable cause.	
7	Work through the remaining symptoms — dead pixels, burn-in, wrong colours, no image with a lit	

Step	What you did	What you observed
	backlight, and intermittent image when the lid moves — and assign a component to each.	
8	For the 'intermittent image when the lid moves' symptom, note the display cable running through the hinge as the classic cause and add a physical inspection step.	
9	Complete the decision table with a column for the confirming test you would run before ordering a replacement part.	
10	Write a customer-facing explanation of one fault in plain language with no jargon, suitable for reading aloud to a non-technical user.	

Verification

Success criterion: Your decision table covers all eight symptoms with a named component and a confirming test for each, and correctly separates the backlight, inverter, panel and digitizer as four distinct failure points.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
